

Unit 6
JOINT COMPACTION

Creating quality joints requires the best efforts of both the paving and the compaction crews. Make sure your team holds up its end of the job.



A well-built and properly compacted transverse joint should be flat and smooth.

There are two types of joints, longitudinal and transverse. Longitudinal joints are formed at the parallel intersection of two layers of asphalt. Longitudinal joints can exist between a hot and a cold layer, a hot and a warm layer, or two hot layers laid down simultaneously.

Transverse joints are formed at the perpendicular intersection of two layers of asphalt. Most commonly, a transverse joint is created when paving starts as a continuation of a previously laid mat. The first part of Unit 6 will deal with construction and compaction of transverse joints.

[TRANSVERSE JOINT COMPACTION PATTERNS]

A transverse joint is created when paving begins at a point where the fresh asphalt layer meets a previously laid and compacted asphalt layer. The transverse joint will then be perpendicular to the direction of paving and the direction of compaction.

There are several techniques used to compact transverse joints, but the goal is always the same. The joint should be compacted flat and the area in front of the joint should be flat and smooth without high spots or depressions. Before paving and compacting a transverse joint, there are fundamental best practices that the crew should follow.

First, always make sure the transverse joint is in good condition prior to paving and compaction. Never start with a rounded or irregular edge at the transverse joint.

A utility cold planer, a skid steer with a cold planer attachment, or a circular saw must be used to cut a straight vertical face at the transverse joint. The area where the joint is cut should be the correct thickness and parallel to the line of paving.

User Tip: *When paving a lane and preparing to stop for the end of a shift, the paving crew commonly changes the feeder system control to the manual mode in order to use all the mix and avoid leaving a large pile of mix when the screed is picked up. Consequently, the head of material in front of the screed starts to change and the mat thickness becomes variable. Caterpillar recommends that the crew mark the mat as soon as they switch to manual feeder system control. The transverse joint should be cut at the point where the mat is marked to avoid cutting in a location where the mat is decreasing or increasing in depth.*



Example of a poor starting reference. Rounded face with deep plunge cut will hinder compaction.



Cutting or milling a straight vertical face is a requirement for a transverse joint.

A good starting joint will have a vertical face and the asphalt layer will be flat, not rounded, tilted up or tilted down. The face of the joint should be coated with tack to help create a bond between the cold asphalt layer and the hot asphalt layer. Clean the area of the cold mat just behind the joint so the height reference for the paver screed is accurate.

Another important factor in constructing and compacting a transverse joint is putting the correct starter boards under the paver screed as the screed is set down at the starting point. The starter boards provide the pre-compaction thickness of the mat as the paving crew pulls the screed off the starting joint. For estimating purposes, you can assume that the mat laid by a vibratory screed will compact at a rate of about 6 mm (1/4") per 25 mm (1") of loose mat thickness. Therefore, if the loose depth is 50 mm (2"), the starter boards would need to be 12 mm (1/2") thick. If the paver has a tamping screed, the compaction rate will be much less, typically around 10% and the starter board thickness will be less.



Example of a good saw-cut joint that has been tacked prior to the start of paving.



Checking the thickness of the starting joint helps the crew select the correct starter board thickness.



After the paver pulls off the transverse starting joint, minimal handwork should be needed.

If the paving crew has done a good job building the transverse starting joint, the joint should need only minor handwork. If the joint is too high or too low, substantial handwork may have to be done prior to the start of the joint compaction process. Joint compaction should begin as soon as all corrections, if any, have been completed.

The transverse joint compaction technique recommended by Caterpillar is designed to flatten the hot / cold transverse joint while maintaining the smoothness of the mat in front of the joint.





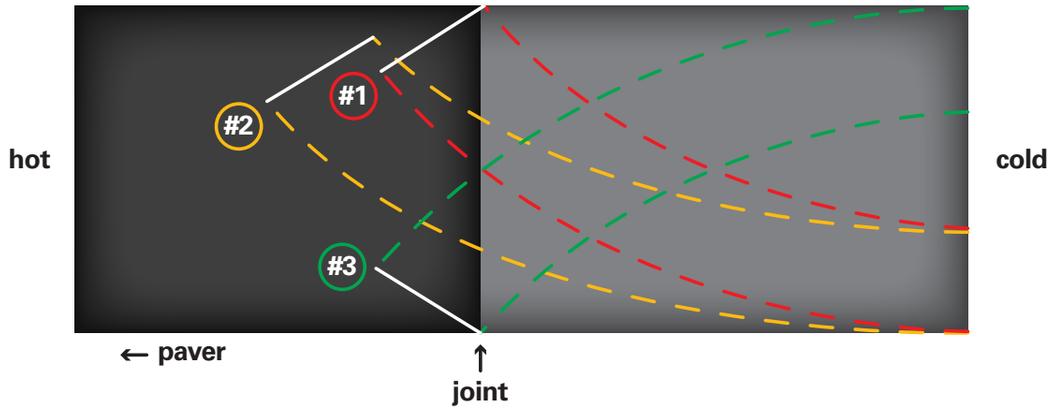


User Tip: *Caterpillar recommends that the crew measure the height of the compacted starting joint. Then, subtract that number from the thickness of the loose mat that was being laid during the previous shift. The result will be an accurate starter board thickness. For example, if the compacted transverse joint height is 40 mm (1.6") and the thickness of the loose mat laid by the screed was 50 mm (2"), then the starter boards should be as close to 10 mm (0.4") as possible.*

The compactor operator can start in the center of the cold compacted mat or along one side of the compacted mat. Proceed forward, starting to turn the drums at an angle as the first drum approaches the joint. Work the front drum across the outer portion of the joint at an angle with both drums static, being careful not to distort the edge of the fresh mat if the edge is unconfined. Back up in the same path.

Move over to one edge of the cold, compacted mat. Proceed forward in the static mode and work the front drum across the joint in the center of the mat. Reverse in the same path.

Move to the center of the cold mat. Proceed forward and work the front drum at an angle across the remaining outer portion of the mat. Use a straight edge to verify that the joint has been compacted flat across its entire width. Repeat static passes if needed.



This transverse joint rolling pattern provides two benefits. First, the approach to the hot / cold transverse joint is at an angle. The angled approach helps flatten the hot mix while minimizing the tendency for the drum to push mix away from the joint.

Second, all the drum stop marks on the fresh mat in front of the joint are left at an angle to the direction of compaction. As the initial phase compactor starts its first pattern, the stop marks will be cleaned up and smoothness will be better in the joint area. Notice, also, how the operator made sure to pull farther forward at the end of the second pass so stop marks were not left in the same area.



Initial compactor starting first pattern and cleaning up stop marks left during joint compaction pattern.

When all the elements of paving and compacting the transverse joint have been done correctly, the operator of the initial compactor should be able to start the established rolling pattern.

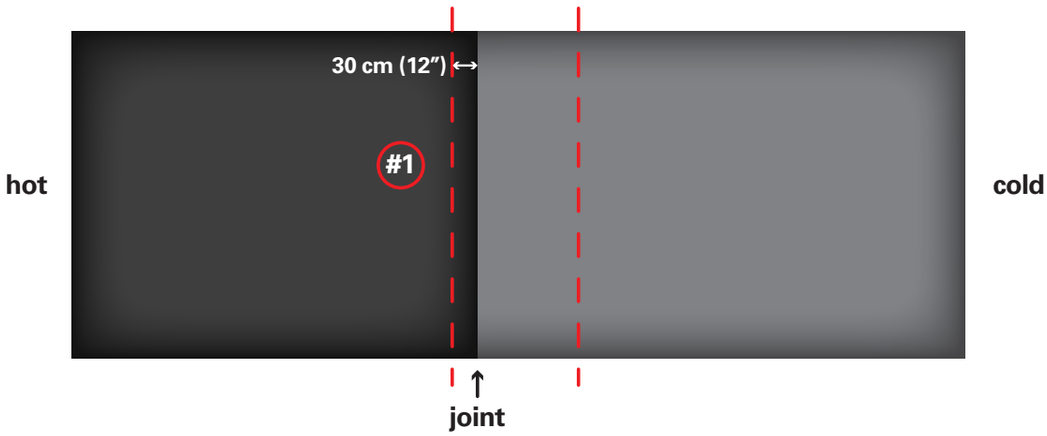
The paver should not have to wait for the transverse joint to be compacted, but should be able to pave at the calculated speed and still be a reasonable distance away from the joint as initial compaction starts.

Another transverse joint rolling pattern that is sometimes used requires adequate space for the compactor to approach the joint from the side.

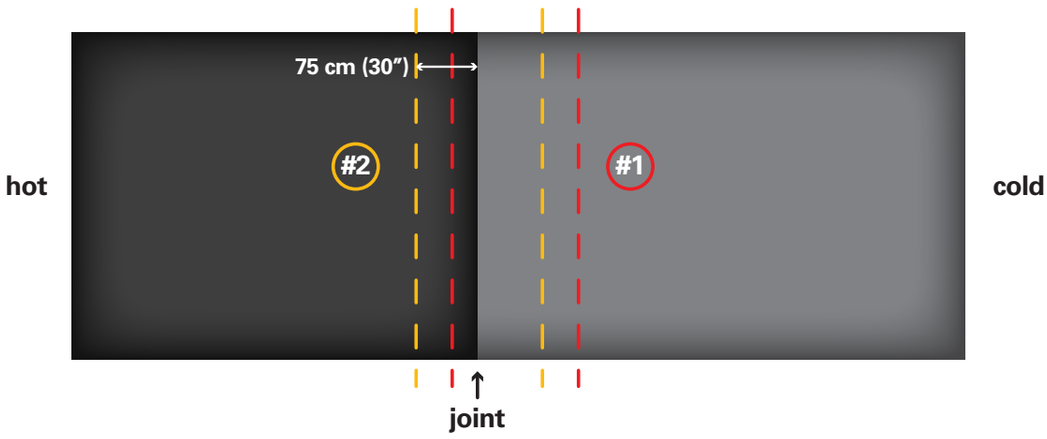
Some crews prefer, when possible, to compact a transverse joint by rolling across the joint from the side. This technique is very effective at flattening the joint, but it leaves drum edge cut marks that are perpendicular to the direction of compaction.



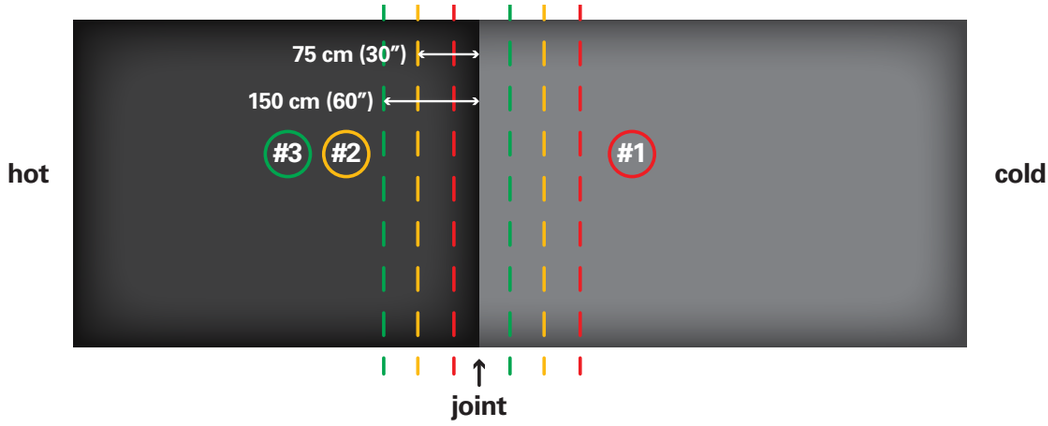
When space permits, compacting a transverse joint from the side can be done.



When compacting a transverse joint from the side, make the first pass with most of the drums on the cold mat with a 30 cm (12") overlap on the hot mat. Check the flatness of the joint.



If another pass is necessary to flatten the transverse joint, move over with more of the drums on the hot mat. The larger overlap will help clean up the first drum edge mark.



If another pass is necessary, move across the joint with most of the drum on the hot mat to clean up the second drum edge mark. The drum edge cut mark left by the third pass is perpendicular to the direction of compaction. When the initial compactor starts its first pattern, it will tend to push over the drum edge mark and can create a bump a short distance in front of the transverse joint. Caterpillar does not recommend this pattern for any project that will be measured for smoothness.

All transverse joint passes, no matter what pattern is used, should be made in static mode. It should not be necessary to activate the drum vibratory system in order to flatten the joint. If a utility compactor is used for transverse joint compaction, low amplitude vibration may be used, if necessary.

If a joint is constructed poorly, it may require a lot of handwork before compaction can begin. Never use the compactor to try to flatten a joint that has the hot mat laid too thickly by the paver.

Remember, the mat will only compact so much. If you continue to vibrate on the joint, you will fracture aggregates, overwork the mat in that area and end up losing density due to over-compaction.

Finally, Caterpillar does not recommend compacting a transverse mat by rolling straight across the joint from the cold side to the hot side.

If a compactor rolls straight across the transverse joint, the drum tends to bump over the hot mat rather than tuck the mix under the drum. Also, the drum is moving mix away from the face of the joint, an action that can contribute to high air voids and premature joint failure.

Summary: *Transverse joint compaction is truly a team effort. First, the paving crew must construct the joint correctly, leaving the correct pre-compaction height and a flat surface without bumps or low spots. Then, the compaction crew must flatten and seal the joint without distorting the mat or creating bumps. Adherence to best practices is the key to paving and compacting transverse joints.*

[LONGITUDINAL JOINT COMPACTION PATTERNS]

Longitudinal joint construction and compaction involve another multi-step process that requires doing many fundamentals correctly. How the compaction team approaches the work on longitudinal joints depends on the objectives of the project.

If appearance is the main objective, then the compaction process should focus on making the joint as invisible as possible. Joint appearance is normally most important for parking lots and city streets.

If joint density is the main objective, then the compaction process should focus on achieving high density at and around the longitudinal joint. Joint density is normally most important on high-volume roads and airports. The process of creating the highest density at the longitudinal joint is the first topic to be addressed.



Having a straight edge for joint matching is a key part of building a quality longitudinal joint.

User Tip: *A useful option for some applications is an edge cutter. The edge cutter installs on Cat Asphalt Compactors in order to trim unconfined edges. The trimmed edges provide an improved vertical face and a better line for matching.*

[HIGH-DENSITY LONGITUDINAL JOINTS]

The first step in creating a quality, high-density longitudinal joint is building it correctly during the paving process. The paver operator should have a steering guide, paint stripe or string to follow. The joint edge should be as straight as possible in order to make the joint matching as easy as possible.

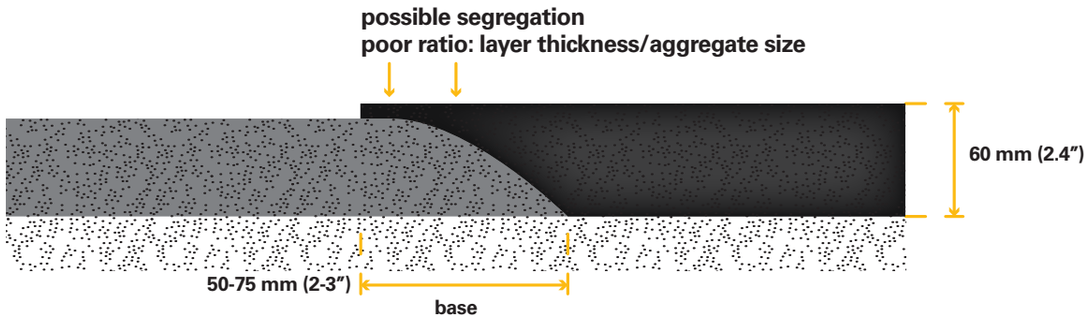
Next, the paving crew should have the end gate ski in contact with the grade being paved. The end gate ski should float on the grade and create a uniform vertical edge that will provide a good bonding surface for joint matching.





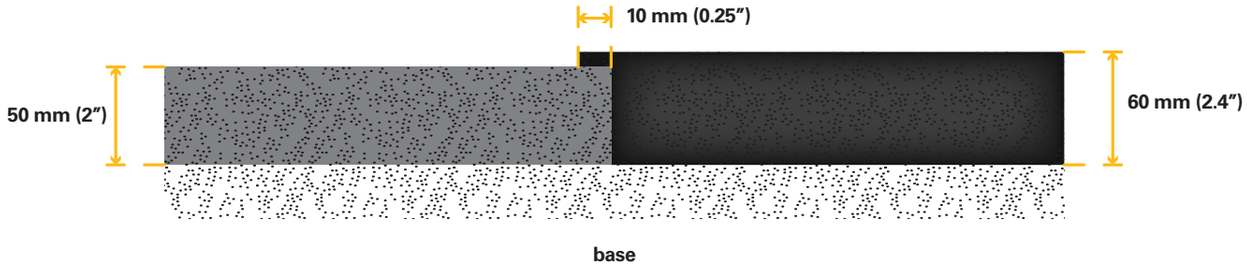
Create a vertical unconfined edge by running the screed end gate in the float position on the grade being paved.

INCORRECT SQUARE JOINT — END GATE UP



When the paving crew operates the screed with the end gates in the raised position, the unconfined edge rolls over, especially when being compacted. The sloped edge causes larger aggregate to drag under the screed when the joint is being matched by the next paving operation. You are likely to see broken aggregate along the longitudinal joint when the joint is compacted. Caterpillar recommends always running the end gates down in the float position when creating an unconfined edge that will be matched by the next lane.

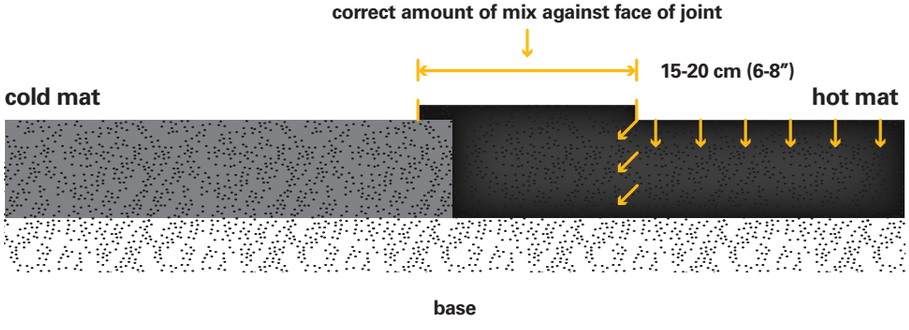
CORRECT SQUARE JOINT — END GATE DOWN



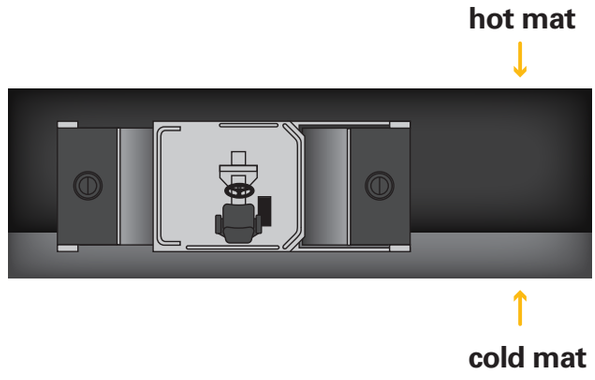
Finally, when the paving crew matches the unconfined edge to create the longitudinal joint, they should overlap the cold asphalt layer about 10 mm (0.25"). The overlap is necessary to make sure there is enough material at the joint to provide a good seal in order to prevent moisture penetration. The height of the hot asphalt layer should be enough to allow for its compaction rate. In the example above, the cold compacted layer is 50 mm (2") thick. The hot asphalt layer is laid 60 mm (2.4"). After compaction, the hot asphalt layer will match the height of the cold mat, assuming that the crew has correctly calculated the compaction rate. Remember, the compaction rate, as a general rule, is about 6 mm (1/4") per 25 mm (1") of screed laid thickness when a vibratory screed is used and about 5 mm (1/5") per 25 mm (1") when a tamping and vibrating screed is used. Always verify the compaction rate of the fresh layer when building a longitudinal joint.

Longitudinal joint raking should be avoided. If the joint overlap and the height of the hot layer are correct, the joint will not need to be raked. Light bumping or minor handwork is permissible. The paving crew should immediately correct the paving technique if excessive raking is needed prior to compaction.

FIRST PASS FOR HIGHER JOINT DENSITY



When joint density is the main objective of the compaction process, the first pass made by the initial phase compactor should be made with both drums on the hot mat about 15-20 cm (6-8") away from the joint. By keeping the drums slightly away from the hot / cold joint, asphalt mix is pushed toward the vertical joint face. Pushing mix toward the joint helps ensure that there will be fewer air voids in the mat after compaction is completed.



During the return pass along the longitudinal joint, the drums should slightly overlap onto the cold mat. The slight overlap will begin the process of creating joint density, sealing the joint and knocking down the hot mat so its height will be the same as the cold mat.



Pneumatic compactor sealing a hot / cold longitudinal joint.

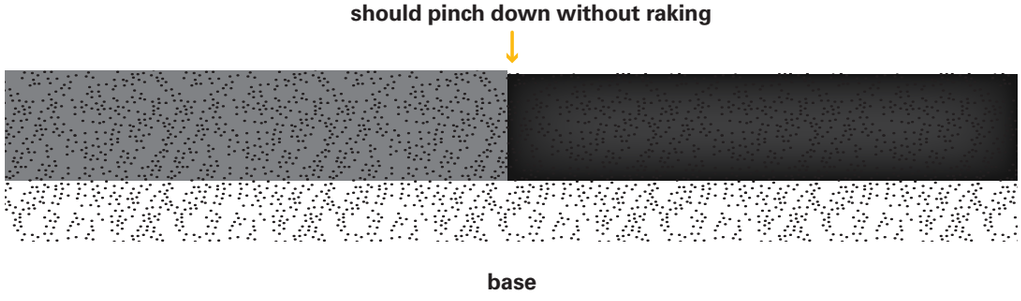
During all phases of compaction and with all types of compaction equipment, the longitudinal joint can be overlapped once the initial pass has been made.

Pneumatic compactors are especially good at knocking down the hot layer in order to equalize the height of the two asphalt layers.

User Tip: *If a vibratory compactor is used to overlap the joint, be careful not to overlap more than about 16 cm (6"). If too much of the vibratory drum is on the cold, compacted mat, drum bouncing is likely to occur. You may see ripples in the fresh mat and fractured aggregates showing on the cold side of the joint.*

User Tip: *If there is a slope difference between the cold mat and the hot mat, be careful not to straddle the joint. Two different slopes create a crown along the joint. The crown is specified for drainage. Straddling the joint will knock down the crown and interfere with moisture drainage.*

CORRECT SQUARE JOINT — END GATE DOWN



The compacted longitudinal joint should result in height match between the two layers and a high degree of density. The joint should be sealed and tight to resist moisture penetration. Remember, there are three keys to achieving good longitudinal joint density.

- *Joint faces are vertical with the correct pre-compaction height*
- *No raking—light bumping permitted*
- *First pass 15-20 cm (6 -8") away from the joint*

[JOINT COMPACTION FOR APPEARANCE]

On some projects, the main goal is to make the longitudinal joint disappear as much as possible. The task is made easier if the longitudinal joint is between hot mats being laid down simultaneously by multiple pavers working in echelon. Creating better appearance is also easier if the joint is between a hot mat and a warm mat that is still somewhat pliable at the surface.



Sealing the longitudinal joint during the first pass improves the appearance of the joint.

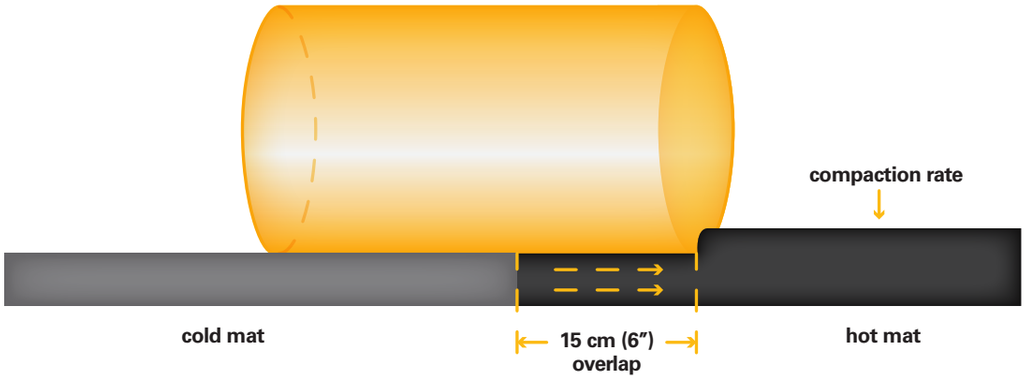
To create a longitudinal joint with the best final appearance, make the first pass along the joint with most of the drums on the cold side of the joint and a slight overlap on the hot side. The compactor must be operated in the static mode during this pass to avoid bouncing on the cold side.

Before setting up a pattern to include a first pass to seal the joint, be sure you have considered the following factors:

- Is there enough time to include this pass in the pattern? Since you make this pass in the static mode with the drums mostly on the cold mat, there will be no density increase in the hot mat. You may have to increase compactor working speed in order to keep up with the paver. Use the Cat Interactive Production Calculator to verify that the extra pass is possible.
- How much mat temperature will be lost by adding the extra pass? The temperature of the asphalt layer is critical to achieving target density. When the ambient temperature is low and the mat is thin, heat loss occurs quickly. It may not be possible to use the initial phase compactor for the pass that seals the joint. Perhaps you'll need to add another compactor to complete this extra pass.
- Is there enough space on the cold side of the joint to accommodate the width of the compactor? If you are working on a road or street project, there may be traffic cones or some sort of barrier close to the longitudinal joint that would prevent you from working on the cold side. You may have to use a narrow drum compactor more suitable for operating in a confined space.

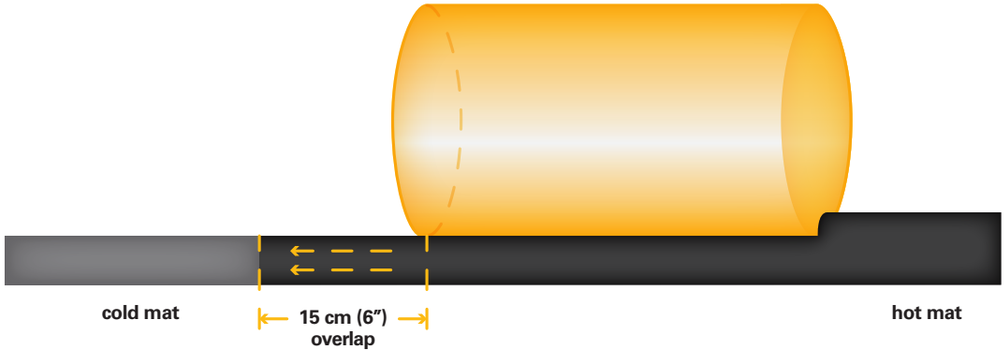
If you're paving and compacting a parking lot or working in a new residential area, you probably will not have to worry about traffic or space. However, you should always confirm production requirements and mat temperatures when planning to include a joint-sealing pass.

LONGITUDINAL JOINT — FIRST PASS



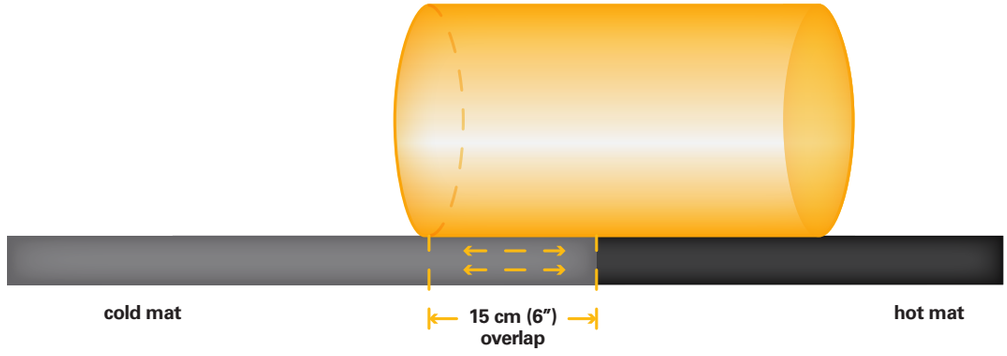
During Pass One, the drums slightly overlap onto the hot side. From an appearance standpoint, the drums effectively push the hot mix down to make the joint height equal on both sides. From a density standpoint, some mix is pushed away from the joint as there is no confinement near the drum edge.

LONGITUDINAL JOINT — SECOND PASS



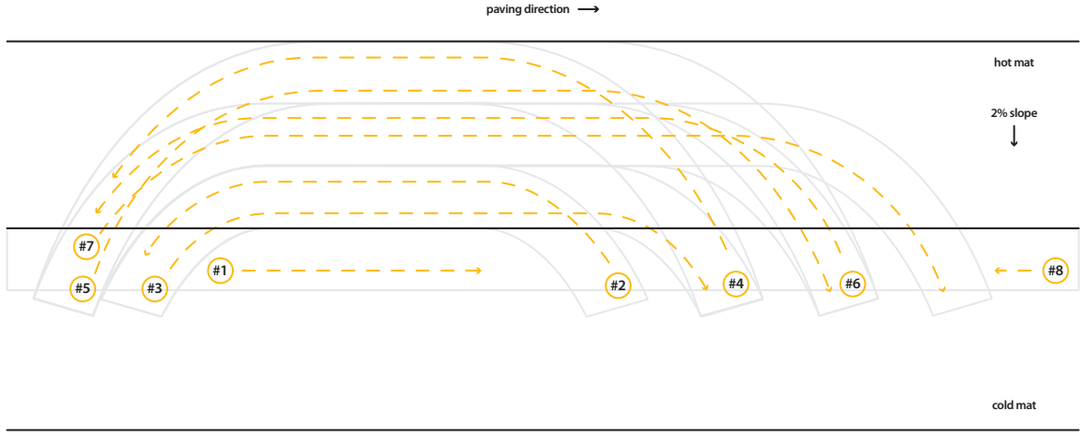
During Pass Two, position the compactor with both vibratory drums completely on the hot mat with the drum edge about 15 cm (6") away from the joint face. This vibratory pass starts to create the required density and tends to push a little mix back toward the longitudinal joint.

LONGITUDINAL JOINT — THIRD PASS



During Pass Three, position the compactor so the drums slightly overlap the longitudinal joint with most of the drums on the hot mat. Since the overlap onto the cold mat is small, you can operate with the drums vibrating. All other passes, if any, with all other compactors can overlap the joint during passes adjacent to the longitudinal joint.

COMPLETE ROLLING PATTERN



Assuming that two passes achieve target density, the rolling pattern looks a little different because of the first pass to seal the joint. In this pattern, you also can take advantage of the cold mat to stop and reverse the compactor. Notice that during Passes Four and Five the operator has moved to the unconfined edge. The unconfined edge should be compacted after the joint when the mat slopes down from the joint toward the unconfined edge. Passes Six and Seven complete the mat coverage by compacting the center of the mat.

At the end of Pass Seven, the compactor operator has stopped to reverse on the cold mat and is located some distance ahead of the point where the joint has already been sealed. Therefore, Pass Eight is a return static pass with most of the drums on the cold mat. After Pass Eight, the operator can position the compactor to begin a new pattern. Because of the joint-sealing pass, the rolling patterns end with the initial compactor in different locations.

If you use a rolling pattern like the one shown, the pass counting map is a good option to help the operator develop consistency.

[COMPACTION OF SPECIALTY JOINTS]

In addition to vertically faced joints, there are wedge joints and notched wedge joints. These types of joints may be specified on highway projects for traffic safety.

Some public works departments require the construction of a notched wedge joint whenever there is a possibility that an unconfined edge may be opened to traffic and that unconfined edge is 50 mm (2") high or higher. The purpose of the notched wedge is to make it easier for vehicles to cross the open vertical edge.

As an aid to the compaction of notched wedge joints, towed rollers are sometimes attached to the paver screed. The towed roller is normally used when the asphalt layer includes a notch that is at least 50 mm (2") high and a wedge that is at least 50 mm (2") thick. Properly constructing notched wedges is the key to achieving good notched wedge joint density.

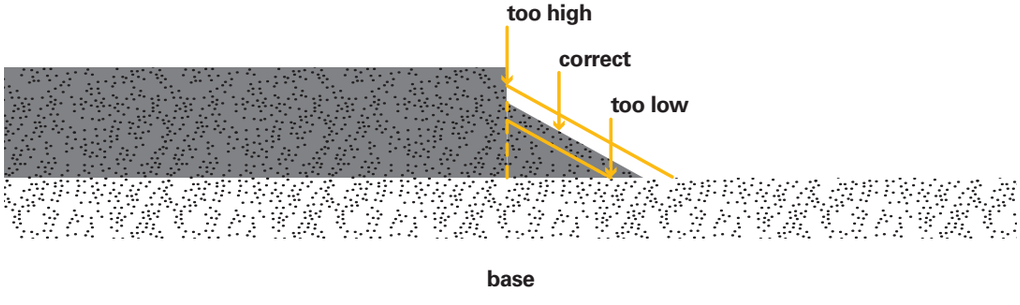


A notched wedge joint being laid down by the paver.



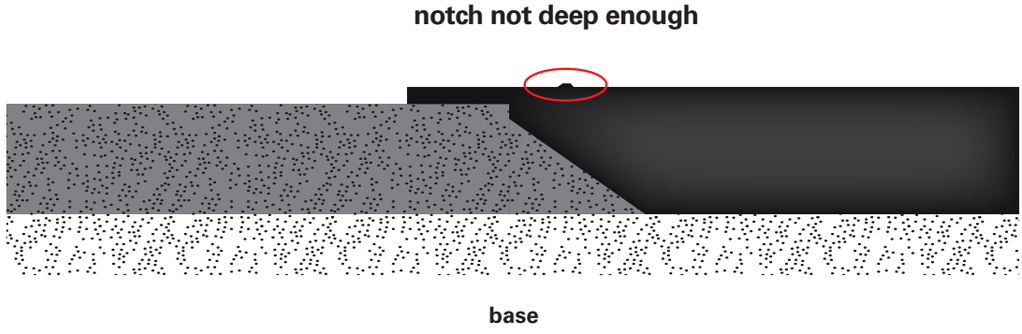
Towed rollers may be used in conjunction with notched wedge joints.

NOTCH WEDGE



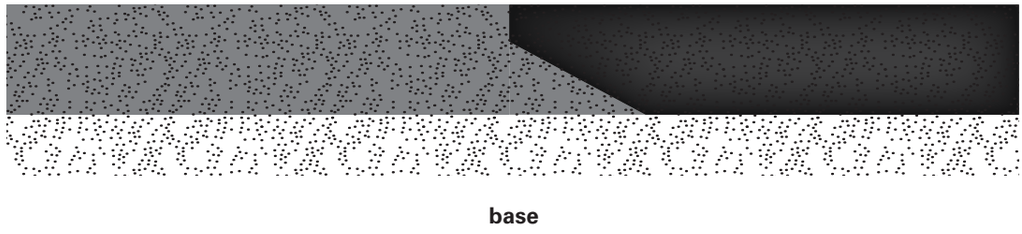
Planning for the correct notch height and wedge thickness is critical when paving notched wedges. The height of the notch should be at least twice the size of the largest aggregate in the asphalt mix. Likewise, the thickness of the wedge should be at least twice the size of the largest aggregate. If the notch is too short, aggregate will drag along the face of the notch. If the wedge is too thin, aggregate will drag along the edge of the wedge.

INCORRECT NOTCH WEDGE



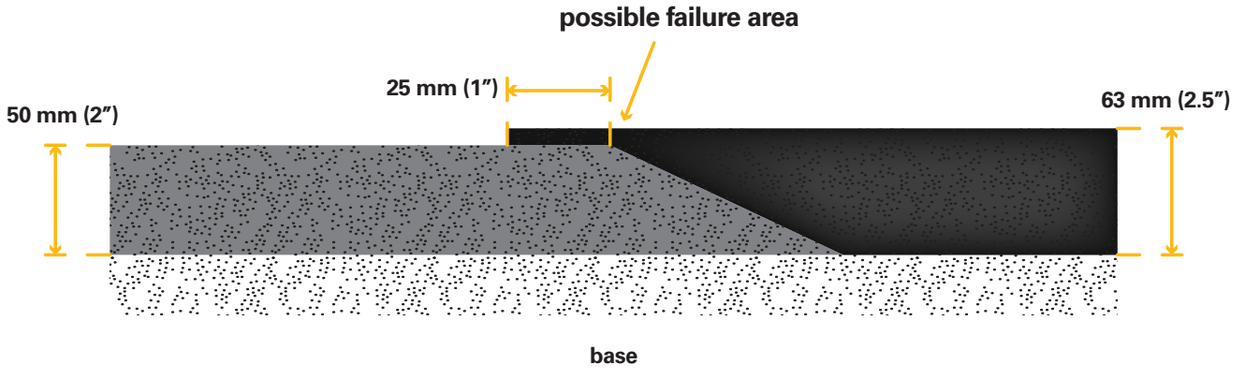
A shallow notch will result in a line of segregated large aggregate along the face of the notch. You may see uncoated rock surfaces in this area, a sign that aggregates are being fractured because the layer is too thin. Over time, moisture will penetrate through the segregated material and the joint will begin to separate, causing premature joint failure.

CORRECT NOTCH WEDGE

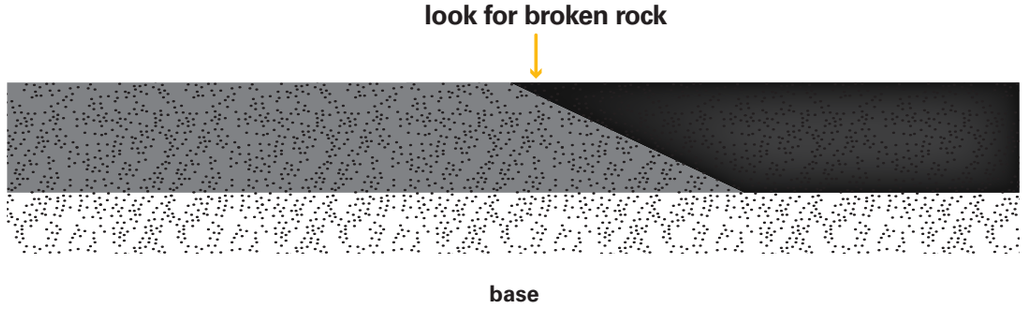


When the notched wedge is built correctly, it can be compacted correctly. Follow the same procedures as if you were compacting a joint with a vertical face. Stay 15-20 cm (6-8") away from the joint with the drums entirely on the hot mat during the first pass along the longitudinal joint. Most research shows that notched wedge joints have as high or even higher density when compared to the densities of vertical joints. Wedge joints are another option sometimes specified by public works departments for the same reason that notched wedges are specified: traffic safety.

WEDGE JOINT



WEDGE JOINT



Wedge joints have an inherent problem due to lack of mat thickness in the area next to the intersection of the hot and cold mats. There is no vertical notch. There is only an angled face, or wedge, upon which the fresh asphalt is placed. Therefore, there is always a possibility of segregation at the top of the wedge. Prior to compaction, you may see a segregation stripe just inside of the joint. After compaction, you may see a stripe of uncoated rock just inside the joint. Wedge joints are most suitable when the largest aggregate in the mix formula is 9 mm (3/8"). Mixes with larger aggregates are prone to segregation at the wedge joint.

Summary: *Much like transverse joints, building quality longitudinal joints requires the best efforts of both the paving crew and the compaction crew. The compaction process cannot correct mistakes made during the paving process. When troubleshooting longitudinal joint problems, start by looking at the edge of cold mats. Then verify that overlap and height of the hot mat are correct. Finally, adjust your rolling pattern to conform to the requirements for joint density or for joint appearance.*


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Unit 7
COMPACTION ISSUES

Compaction issues typically result from variability. Start your troubleshooting efforts by looking for variability in the paving process, mat temperature, rolling patterns, and rolling speed.



Stone Matrix Asphalt with polymer-modified asphalt cement stuck to steel drum.

This unit reviews some of the most common compaction issues that are faced by operators, quality control personnel and supervisors. There is no significance to the order in which the issues are presented. When possible, similar issues are grouped for ease of understanding.

[ASPHALT PICK-UP ON DRY DRUM SURFACES]

The most common cause of downtime on double drum, asphalt compactors is a malfunctioning drum spray system.

If any part of the steel drum does not have a water film on the surface, hot asphalt is likely to stick to the drum. The stickier the asphalt mix, the more severe the problem. A small amount of asphalt pick-up on the drum quickly becomes a large problem. With each drum rotation, the amount of pick-up increases and the mat will begin to show divots.



Divot made in the fresh mat by asphalt stuck to the drum.



Water spray nozzles operate properly when maintenance guidelines are followed and clean water is used.

When asphalt begins to stick to a drum surface, operation of that compactor must be terminated until the drum is completely clean and any problem with the drum spray system is repaired. Continuing to operate the compactor will result in severe mat damage that will require extensive handwork to fill in and level divots left in the mat.

- **Use clean water.** Whenever possible, fill water spray reservoirs with water from approved sources. If you have to use pond water, for example, increase the frequency of maintenance steps.
- **Change main spray system filters.** Follow the filter change interval shown in the machine's Operation and Maintenance Manual. When the main spray system filter is plugged, water bypasses the filter and unfiltered water goes to the spray bars. Unfiltered water is more likely to cause spray nozzles to plug. Always have a spare filter stored on the compactor or in the maintenance vehicle.
- **Maintain inlet filters.** Most water reservoirs have an inlet filter inside the reservoir fill port. The inlet filter is the first stage of water filtration. Do not discard the inlet filter. Place the water supply hose inside the inlet filter.
- **Clean spray nozzles.** The spray nozzles have internal brass or plastic screens. Nozzle screens should be examined daily for contamination. Clean nozzle screens thoroughly as needed. If you use contaminated water, increase the frequency of nozzle maintenance. If only one side of the nozzle is plugged, the spray pattern will be smaller and can cause the dry strip on the drum and start the asphalt pickup.

The primary cause of dry areas on the drum surface is a plugged spray nozzle. Good maintenance and clean water supply are the keys to preventing plugged spray nozzles.

- **Maintain water distribution mats.** The drum will have some type of water distribution mat to help spread the water film evenly on the drum surface. As the distribution mats wear, you may have to adjust them to maintain good drum contact. Replace distribution mats according to wear indicators.
- **Understand spray system capabilities.** Most water spray systems offer full-time or intermittent spray. Never sacrifice water coverage in an effort to conserve water. It is better to stop more frequently for water refills than to stop for drum cleaning.
- **Protect the water spray system during cold weather.** An optional water spray system antifreeze kit is available. It includes a separate reservoir for antifreeze. At the end of the shift, the operator can circulate antifreeze through the system to prevent overnight freezing.

User Tip: *Know how to operate the water spray system in the event of water spray pump failure. Most water spray systems include two water pumps and most systems have the capability to supply both spray bars with only one pump. Know what needs to be done to operate with one pump while a replacement pump is being delivered.*



Hot asphalt can stick to rubber tires and the clumps can fall off the tires to create a substantial loss of smoothness and cosmetic appearance.

[ASPHALT PICK-UP ON RUBBER TIRES]

Asphalt also can stick to rubber tires. The severity of asphalt sticking to rubber tires depends primarily on the stickiness of the asphalt. Tire pick-up is also affected by the difference in temperature between the surface of the asphalt layer and the rubber tires.

When asphalt begins to stick to the rubber tires of a pneumatic compactor, the operator must immediately stop and correct the problem.

- Use a bio-degradeable release agent to clean the affected tires. Apply more release agent to the tires before resuming the compaction process.
- Be sure the distribution mats and tire scrapers are properly positioned and in good working condition.
- Move the pneumatic compactor on the asphalt layer in an area where the surface temperature is relatively low.
- Heat the tires by operating on the warm mat before moving ahead to a higher temperature zone.





Wheel covers are an important feature on pneumatic compactors and should be installed to help prevent heat loss in the rubber tires.

Heating the rubber tires and keeping them at the correct temperature are very important. Wheel covers help keep heat confined around the front and rear axles. Caterpillar recommends using wheel covers on pneumatic compactors for all asphalt compaction applications. Covers are especially important when compacting asphalt that contains modified asphalt cement. If wheel covers are not installed, the tires are exposed to the ambient conditions and can lose heat rapidly.

Release agents are sometimes used to help prevent hot asphalt from sticking to rubber tires. Always confirm with the public works department what release agents are permitted.

Most compactors have a tire spray system that is often filled with water and an additive. Common additives include detergents, water softeners or purpose-designed additives that increase the film thickness of the water sprayed on the tires.

User Tip: *When loading a pneumatic compactor on a transport, be sure to roll up the wheel covers and secure them in the transport position. If you leave the covers down, the tires may run over the covers and damage them.*